COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	8/27/2026	**DATE SUBMITTED:**	8/27/2026
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ACTIVITY NO. 5

Implement Ansible roles in playbooks

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (CPE232_yourname). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issue this command. Did something happen? Why?

```
quioyo3@workstation:~/CPE2112_quioyo$ git pull
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0
Unpacking objects: 100% (5/5), 777 bytes | 777.00 KiB/s, done
From github.com:Quioyo/CPE212_quioyo
   4a3e529..1752c44  main       -> origin/main
Updating 4a3e529..1752c44
Fast-forward
 ansible.cfg        | 11 ++++++++++
 install_apache.yml | 15 +++++++++++++++
 inventory          |  8 ++++++++
 3 files changed, 34 insertions(+)
 create mode 100644 ansible.cfg
 create mode 100644 install_apache.yml
 create mode 100644 inventory
quiioyo3@workstation:~/CPE212_quioyo$ ls
ansible.cfg  install_apache.yml  inventory  README.md
quiioyo3@workstation:~/CPE212_quioyo$
```

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): ansible-playbook --ask-become-pass install_apache.yml. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the

package manager. The default package manager for Centos is "yum."

```
PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
fatal: [centos]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to th
e host via ssh: quioyo1@192.168.56.112: Permission denied (publickey,gsapi-keyex,gs
api-with-mic,password).", "unreachable": true}
ok: [node1]
ok: [node2]

TASK [update repository index] *************************************************
changed: [node1]
changed: [node2]

TASK [install apache2 package] *************************************************
changed: [node1]
changed: [node2]

TASK [add PHP support for apache] **********************************************
changed: [node1]
changed: [node2]
```

3. Edit the install_apache.yml file and insert the lines shown below.

Make sure to save the file and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.
If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.
- name: update repository index
apt:
update_cache: yes
when: ansible_distribution in [“Debian”, “Ubuntu]
Note: This will work also if you try. Notice the changes are highlighted.

```
-
hosts: all
become: true
tasks:

- name: update repository index
  apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache2 package
  apt:
    name: apache2
  when: ansible_distribution == "Ubuntu"

- name: add PHP support for apache
  apt:
    name: libapache2-mod-php
  when: ansible_distribution == "Ubuntu"
```

```
PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
fatal: [centos]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to th
e host via ssh: quioyo1@192.168.56.112: Permission denied (publickey,gssapi-keyex,g
api-with-mic,password).", "unreachable": true}
ok: [node2]
ok: [node1]

TASK [update repository index] *************************************************
changed: [node1]
changed: [node2]

TASK [install apache2 package] *************************************************
ok: [node1]
ok: [node2]

TASK [add PHP support for apache] **********************************************
ok: [node1]
ok: [node2]
```

4. Edit the install_apache.yml file and insert the lines shown below.

Make sure to save and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the Result.

```
-
hosts: all
become: true
tasks:

- name: update repository index
  apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache2 package
  apt:
    name: apache2
    stae: latest
  when: ansible_distribution == "Ubuntu"

- name: add PHP support for apache
  apt:
    name: libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: update repository index
  dnf:
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: install apache2 package
  dnf:
    name: httpd
    state: latest
  when: ansible_distribution == "CentOS"
```

```
skipping: [centos]
ok: [node1]
ok: [node2]

TASK [add PHP support for apache] ************************************************
skipping: [centos]
ok: [node1]
ok: [node2]

TASK [update repository index] ***************************************************
skipping: [node1]
skipping: [node2]
ok: [centos]

TASK [install apache2 package] ***************************************************
skipping: [node1]
skipping: [node2]
ok: [centos]

TASK [add PHP support for apache] ************************************************
skipping: [node1]
skipping: [node2]
changed: [centos]

PLAY RECAP ***********************************************************************
centos                     : ok=4    changed=1    unreachable=0    failed=0    skip
ped=3    rescued=0    ignored=0
node1                      : ok=4    changed=1    unreachable=0    failed=0    skip
ped=3    rescued=0    ignored=0
node2                      : ok=4    changed=1    unreachable=0    failed=0    skip
ped=3    rescued=0    ignored=0
```

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.

5.1 To activate, go to the CentOS VM terminal and enter the following:
systemctl status httpd
The result of this command tells you that the service is inactive.

```
[quioyo@workstation ~]$ sudo systemctl status httpd
o httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled
     Active: inactive (dead)
       Docs: man:httpd.service(8)

Aug 27 09:40:29 localhost.localdomain systemd[1]: Starting The Apach
Aug 27 09:40:29 localhost.localdomain httpd[93033]: AH00558: httpd:
Aug 27 09:40:29 localhost.localdomain httpd[93033]: Server configure
Aug 27 09:40:29 localhost.localdomain systemd[1]: Started The Apache
Aug 27 09:41:51 localhost.localdomain systemd[1]: Stopping The Apach
Aug 27 09:41:52 localhost.localdomain systemd[1]: httpd.service: Dea
Aug 27 09:41:52 localhost.localdomain systemd[1]: Stopped The Apache
```

5.2 Issue the following command to start the service:
sudo systemctl start httpd
(When prompted, enter the sudo password)
sudo firewall-cmd --add-port=80/tcp
(The result should be a success)

```
[quioyo@workstation ~]$ sudo systemctl start httpd

[quioyo@workstation ~]$ sudo firewall-cmd --add-port=80/tcp
success
```

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)

Task 2: Refactoring playbook
This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also makes run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook install_apache.yml. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

Make sure to save the file and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the Result.

2. Edit the playbook install_apache.yml again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidate everything in just 2 plays. This can be done by removing the update repository play and putting the command update_cache: yes below the command state: latest. See below for Reference:

Make sure to save the file and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the Result.

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the apache_package and php_package are variables. The names are arbitrary, which means we can choose different names. We also take out the line when: ansible_distribution. Edit the playbook install_apache.yml again and make sure to follow the below image. Make sure to save the file and exit.

Run ansible-playbook --ask-become-pass install_apache.yml and describe the Result.

4. Unfortunately, task 2.3 was not successful. It's because we need to change something in the inventory file so that the variables we declared will be in place. Edit the inventory file and follow the below configuration:

Make sure to save the inventory file and exit.
Finally, we still have one more thing to change in our install_apache.yml file. In
task 2.3, you may notice that the package is assigned as apt, which will not run in CentOS. Replace the apt with the package. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use apt, and for CentOS it will automatically use dnf. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation: ansible.builtin.package – Generic OS package manager — Ansible Documentation
Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

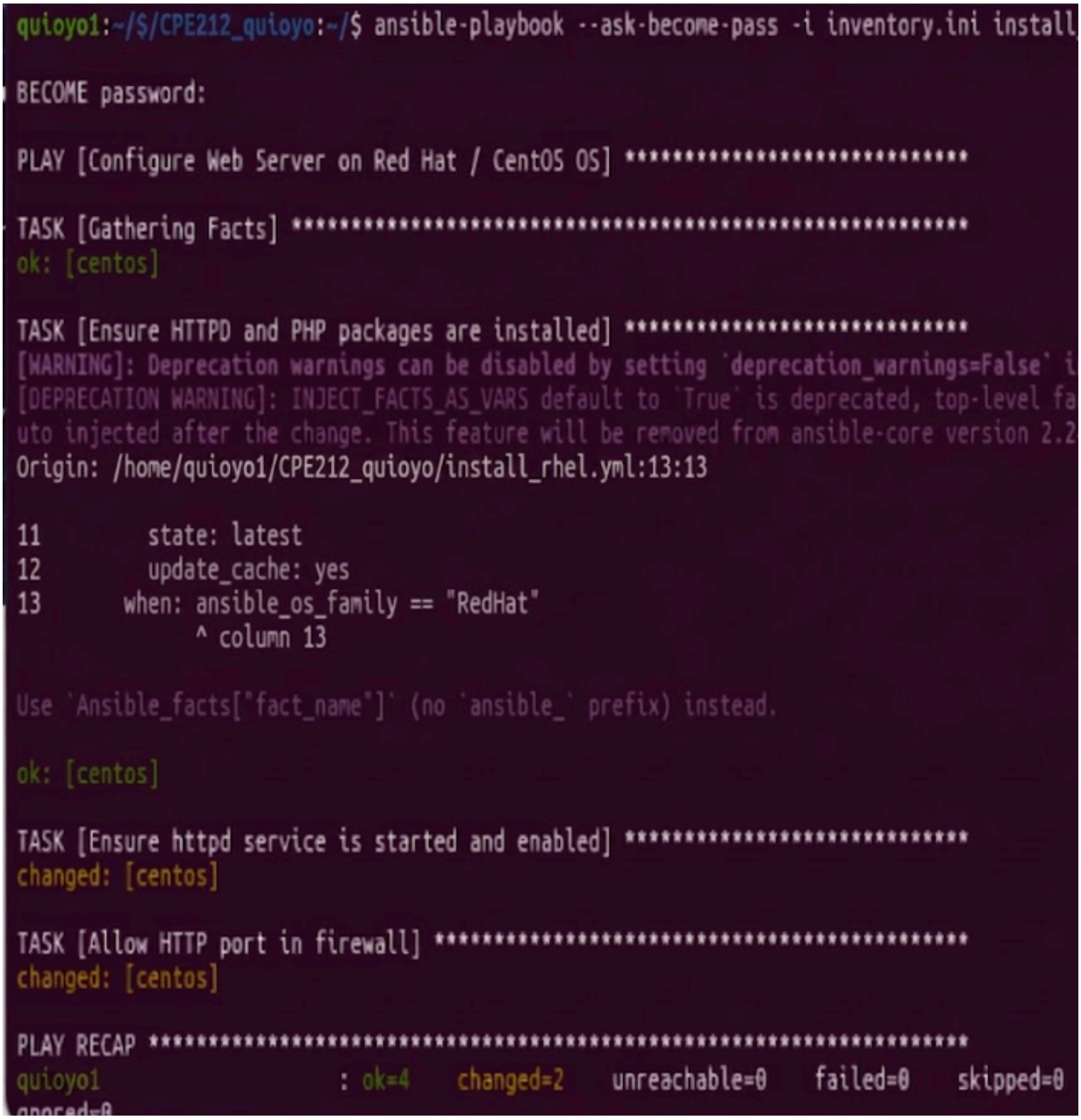
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:
1. Create a playbook that could do the previous tasks in Red Hat OS.

```
quioyo1:~/CPE212_quioyo$ cat install_rhel.yml
---
- name: Configure Web Server on Red Hat / CentO
  hosts: centos
  become: true
  tasks:
    - name: Ensure HTTPD and PHP packages are i
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
      when: ansible_os_family == "RedHat"

    - name: Ensure httpd service is started and
      service:
        name: httpd
        state: started
        enabled: yes

    - name: Allow HTTP port in firewall
      ansible.posix.firewalld:
        port: 80/tcp
        permanent: yes
        state: enabled
        immediate: yes
      ignore_errors: yes
```



1. Why Refactoring Playbook Code Matters

Refactoring removes duplicate tasks and simplifies playbook structure. You reduce errors and make your automation easier to maintain over time.

Simplified code runs faster and uses fewer system resources. You can reuse modular roles across multiple projects to save development time.

2. When to Use the "when" Statement

You use the when statement to run tasks only under specific conditions. You can target specific operating systems like Ubuntu or Red Hat based on host facts.

You can trigger actions based on previous task results or custom variables. This prevents unnecessary changes and keeps your playbooks efficient.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Extra guidance in typing correct codes**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **It was used only for guidance and correct outputs for it to be perfect and what the module said it to be.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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